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(54) **NITRILE-CONTAINING COMPOUNDS USEFUL AS ANTIVIRAL AGENTS FOR THE TREATMENT  
OF A CORONAVIRUS INFECTION**

NITRILHALTIGE VERBINDUNGEN ALS ANTIVIRAL-AGENTEN ZUR BEHANDLUNG VON EINER  
CORONAVIRUS INFEKTION

COMPOSÉS COMPRENANT UN NITRILE COMME ANTIVIRAUX POUR LE TRAITEMENT D'UNE  
INFECTION À CORONAVIRUS

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**WO-A1-2005/113580**

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Viruses 2020, 12, 244; doi:10.3390/v12020244. Very recently, Hilgenfeld and colleagues published a high-resolution X-ray structure of the SARS-CoV-2 coronavirus main protease (3CL) Zhang, L.; Lin, D.; Sun, X.; Rox, K.; Hilgenfeld, R.; X-ray Structure of Main Protease of the Novel Coronavirus SARS-CoV-2 Enables Design of  $\alpha$ -Ketoamide Inhibitors; bioRxiv preprint doi: <https://doi.org/10.1101/2020.02.17.952879>. The structure indicates that there are differences when comparing the 3CL proteases of SARS-CoV-2 and SARS-CoV. In the SARS-CoV but not in the SARS-CoV-2 3CL protease dimer, there is a polar interaction between the two domains III involving a 2.60-Å hydrogen bond between the side-chain hydroxyl groups of residue Thr<sup>285</sup> of each protomer, and supported by a hydrophobic contact between the side-chain of Ile<sup>286</sup> and Thr<sup>285</sup> C<sub>γ2</sub>. In the SARS-CoV-2 3CL, the threonine is replaced by alanine, and the isoleucine by leucine when compared with the same residues in the SARS-CoV 3CL. The Thr285Ala replacement observed in the SARS-CoV-2 3CL protease allows the two domains III to approach each other somewhat closer (the distance between the C $\alpha$  atoms of residues 285 in molecules A and B is 6.77 Å in SARS-CoV 3CL protease and 5.21 Å in SARS-CoV-2 3CL protease and the distance between the centers of mass of the two domains III shrinks from 33.4 Å to 32.1 Å). In the active site of SARS-CoV-2 3CL, Cys<sup>145</sup> and His<sup>41</sup> form a catalytic dyad, which when taken together with a buried water molecule that is hydrogen-bonded to His<sup>41</sup> can be considered to constitute a catalytic triad of the SARS-CoV-2 3CL protease. In view of the ongoing SARS-CoV-2 spread that has caused the current worldwide COVID-19 outbreak, it is desirable to have new methods of inhibiting SARS-CoV-2 viral replication and of treating COVID-19 in patients.

**[0005]** International patent publication WO-2005/113580 describes anti-coronaviral compounds which are inhibitors of the SARS 3C like protease.

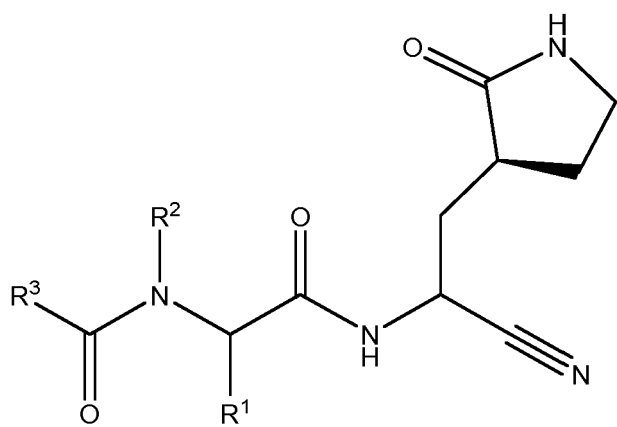
**[0006]** In the article "Inhibition of enterovirus 71 replication by an  $\alpha$ -hydroxy-nitrile derivative NK-1.9k" (Antiviral Research, 141, 2017, 91-100), Wang *et al* describe a specific compound which is said to be a potential treatment for hand-foot-and-mouth disease.

#### Summary of The Invention

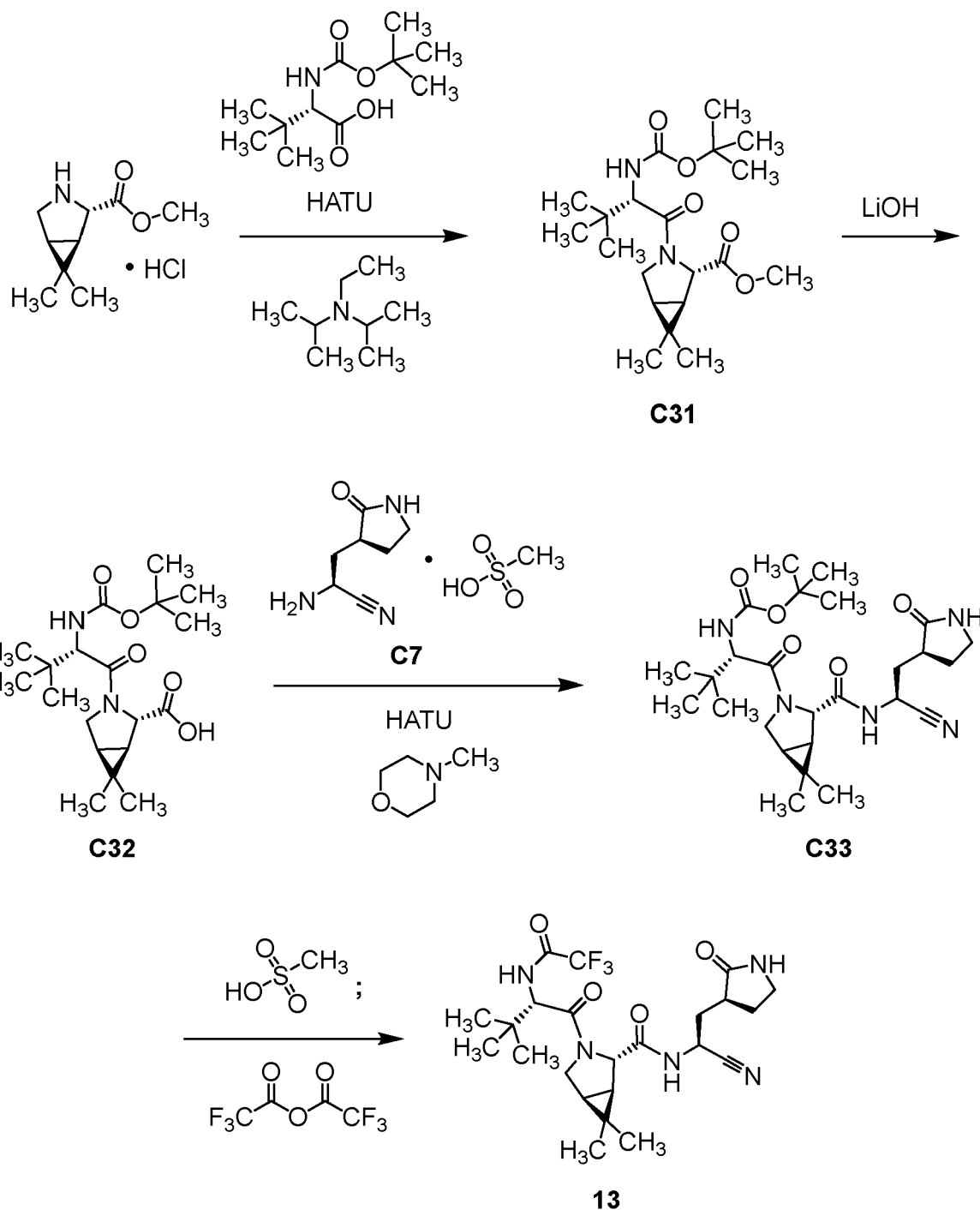
**[0007]** The present invention provides novel compounds which act in inhibiting or preventing SARS-CoV-2 viral replication and thus are useful in the treatment of COVID-19. The present invention also provides pharmaceutical compositions comprising the compounds and also provides the compounds for use in methods of treating COVID-19 and inhibiting SARS-CoV-2 viral replication by administering the compounds of the invention or pharmaceutical compositions comprising the compounds of the invention.

**[0008]** The present invention provides certain compounds, selected from a compound 1 as defined hereafter, falling within the scope of

#### Formula I



and pharmaceutically acceptable salts thereof; wherein R<sup>1</sup> is selected from the group consisting of C<sub>1</sub>-C<sub>6</sub> alkyl which is optionally substituted with a cyano or with one to five fluoro; C<sub>2</sub>-C<sub>6</sub> alkynyl; and (C<sub>3</sub>-C<sub>6</sub> cycloalkyl)-C<sub>1</sub>-C<sub>3</sub> alkyl which is optionally substituted with one to two substituents selected from trifluoromethyl and C<sub>1</sub>-C<sub>3</sub> alkyl or with one to five fluoro; R<sup>2</sup> is hydrogen or R<sup>2</sup> and R<sup>1</sup> taken together with the nitrogen and carbon atoms to which they are attached are a pyrrolidine or piperidine ring which is optionally substituted with one to four R<sup>2a</sup>; R<sup>2a</sup> at each occurrence is independently selected from the group consisting of fluoro, C<sub>1</sub>-C<sub>6</sub> alkyl optionally substituted with one to three fluoro and C<sub>1</sub>-C<sub>6</sub> alkoxy optionally substituted with one to three fluoro; or two R<sup>2a</sup> groups when attached to adjacent carbons and taken together with the carbons to which they are attached are a fused C<sub>3</sub>-C<sub>6</sub> cycloalkyl which is optionally substituted with one to four



Step 1. Synthesis of methyl (1R,2S,5S)-3-[N-(*tert*-butoxycarbonyl)-3-methyl-L-valyl]-6,6-dimethyl-3-azabicyclo[3.1.0]hexane-2-carboxylate (**C31**).

**[0260]** O-(7-Azabenzotriazol-1-yl)-*N,N,N',N'*-tetramethyluronium hexafluorophosphate (HATU; 7.92 g, 20.8 mmol) was added to a 0 °C mixture of *N*-(*tert*-butoxycarbonyl)-3-methyl-L-valine (4.38 g, 18.9 mmol) and methyl (1R,2S,5S)-6,6-dimethyl-3-azabicyclo[3.1.0]hexane-2-carboxylate, hydrochloride salt (3.9 g, 19 mmol) in *N,N*-dimethylformamide (95 mL). After the reaction mixture had been stirred for 5 minutes, *N,N*-diisopropylethylamine (8.25 mL, 47.4 mmol) was added; stirring was continued at 0 °C for 2 hours, whereupon aqueous citric acid solution (1 N, 20 mL) and water (40 mL) were added. The resulting mixture was stirred for 2 minutes, and then diluted with ethyl acetate (250 mL). The organic layer was washed with water (3 × 150 mL) and with saturated aqueous sodium chloride solution, dried over sodium sulfate, filtered, and concentrated *in vacuo*. Purification via silica gel chromatography (Gradient: 0% to 100% ethyl acetate in heptane) afforded **C31** as an oil. Yield: 3.60 g, 9.41 mmol, 50%. <sup>1</sup>H NMR (400 MHz, methanol-*d*<sub>4</sub>) δ 6.42

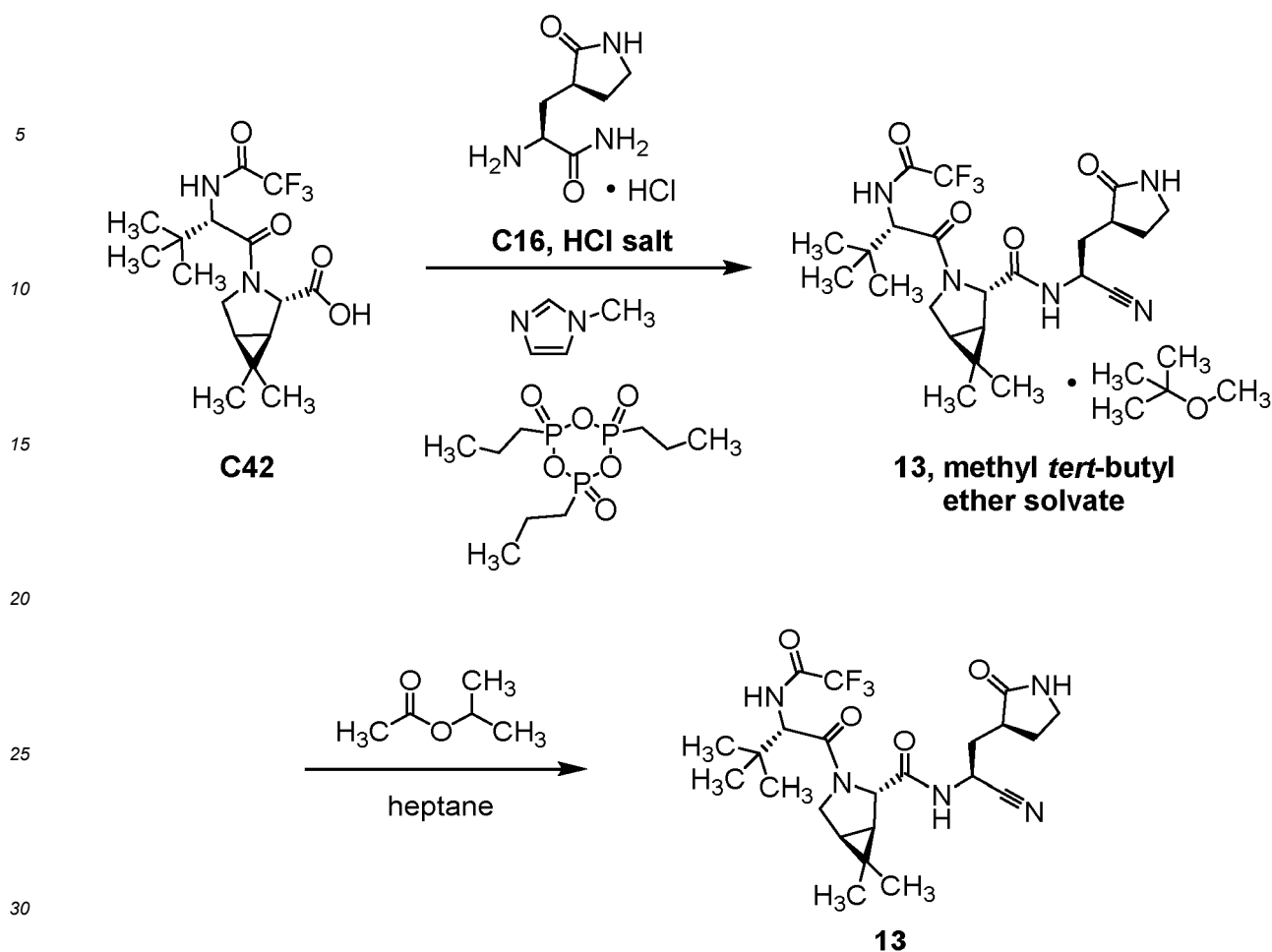
(continued)

| Angle (°2-theta)<br>+/- 0.2° 2-Theta | Relative<br>Intensity (%) | Angle (°2-theta)<br>+/- 0.2° 2-Theta | Relative<br>Intensity (%) | Angle (°2-theta)<br>+/- 0.2° 2-Theta | Relative<br>Intensity (%) |
|--------------------------------------|---------------------------|--------------------------------------|---------------------------|--------------------------------------|---------------------------|
| 11.9                                 | 32                        | 20.9                                 | 12                        | 29.5                                 | 4                         |
| 12.6                                 | 49                        | 21.2                                 | 7                         | 30.3                                 | 3                         |
| 13.0                                 | 4                         | 21.5                                 | 4                         | 30.9                                 | 5                         |
| 14.3                                 | 37                        | 21.8                                 | 3                         | 32.1                                 | 3                         |
| 15.8                                 | 8                         | 22.3                                 | 24                        | 33.4                                 | 5                         |
| 16.1                                 | 22                        | 22.8                                 | 6                         | 33.5                                 | 3                         |
| 16.9                                 | 100                       | 23.0                                 | 9                         | 35.5                                 | 6                         |
| 17.2                                 | 46                        | 23.2                                 | 5                         | 36.5                                 | 3                         |
| 18.6                                 | 42                        | 23.5                                 | 6                         | 38.2                                 | 3                         |
| 18.9                                 | 6                         | 23.8                                 | 17                        |                                      |                           |
| 19.3                                 | 23                        | 25.4                                 | 10                        |                                      |                           |

Third Alternate Synthesis of Example 13

(1*R*,2*S*,5*S*)-*N*-{(1*S*)-1-Cyano-2-[(3*S*)-2-oxopyrrolidin-3-yl]ethyl}-6,6-dimethyl-3-[3-methyl-*N*-(trifluoroacetyl)-L-valyl]-3-azabicyclo[3.1.0]hexane-2-carboxamide (13)

**[0286]**



Step 1. Synthesis of (1R,2S,5S)-N-((1S)-1-cyano-2-[(3S)-2-oxopyrrolidin-3-yl]ethyl)-6,6-dimethyl-3-[3-methyl-N-(trifluoroacetyl)-L-valyl]-3-azabicyclo[3.1.0]hexane-2-carboxamide, methyl tert-butyl ether solvate (**13**, methyl tert-butyl ether solvate)

**[0287]** A 0 °C mixture of **C42** (90.5 mass%, 5.05 g, 12.5 mmol) and **C16, HCl salt** (98.9 mass%, 3.12 g, 14.9 mmol) in acetonitrile (50 mL) was treated with 2,4,6-tripropyl-1,3,5,2,4,6-trioxatriphosphinane 2,4,6-trioxide (50% solution by weight in acetonitrile; 17 mL, 24.3 mmol) over approximately 10 minutes. 1-Methyl-1H-imidazole (4.0 mL, 50.2 mmol) was then added slowly, over approximately 15 minutes, and the reaction mixture was allowed to stir at 0 °C for 3.5 hours, whereupon it was warmed to 25 °C. 2,4,6-Tripropyl-1,3,5,2,4,6-trioxatriphosphinane 2,4,6-trioxide (50% solution by weight in acetonitrile; 17 mL, 24.3 mmol) was added in one portion, and the reaction mixture was stirred at 45 °C for 16 hours. It was cooled to 25 °C at that point, and then treated over 10 minutes with an aqueous solution of sodium bicarbonate (1.14 M; 35 mL, 40 mmol). After addition of ethyl acetate (25 mL) and sufficient water to dissolve the resulting solids, the organic layer was washed twice with an aqueous solution of sodium bicarbonate (1.14 M; 25 mL, 28 mmol). After the organic layer had been washed with aqueous sodium chloride solution (14%, 2 × 20 mL), it was dried over sodium sulfate, filtered, and concentrated in vacuo. The residue was mixed with ethyl acetate (2.1 mL) and treated with methyl tert-butyl ether (19 mL); the resulting slurry was heated with stirring at 50 °C for 1 hour, cooled to 25 °C over 1 hour, and held at 25 °C for 1.5 hours. Solids were isolated via filtration, washed with methyl tert-butyl ether (2 mL/g), and dried in a vacuum oven overnight at 50 °C to afford (1R,2S,5S)-N-((1S)-1-cyano-2-[(3S)-2-oxopyrrolidin-3-yl]ethyl)-6,6-dimethyl-3-[3-methyl-N-(trifluoroacetyl)-L-valyl]-3-azabicyclo[3.1.0]hexane-2-carboxamide, methyl tert-butyl ether solvate (**13**, methyl tert-butyl ether solvate) as a crystalline white solid. The bulk of this material was progressed to the following step. Yield: 3.71 g, 6.31 mmol, 50%. <sup>1</sup>H NMR (400 MHz, DMSO-*d*<sub>6</sub>) δ 9.40 (d, J = 8.4 Hz, 1H), 9.02 (d, J = 8.6 Hz, 1H), 7.66 (s, 1H), 4.97 (ddd, J = 10.7, 8.6, 5.1 Hz, 1H), 4.41 (d, J = 8.4 Hz, 1H), 4.16 (s, 1H), 3.91 (dd, component of ABX system, J = 10.3, 5.5 Hz, 1H), 3.69 (d, half of AB quartet, J = 10.4 Hz, 1H), 3.18 - 3.10 (m, 1H), 3.09 - 2.99 (m, 1H), 2.46 - 2.35 (m, 1H), 2.20 - 2.04 (m, 2H), 1.78 - 1.64 (m, 2H), 1.56 (dd, J = 7.4, 5.6 Hz, 1H), 1.32 (d, half of AB quartet, J = 7.6 Hz, 1H), 1.03 (s, 3H), 0.98 (s, 9H), 0.85 (s, 3H); methyl tert-butyl ether peaks: 3.07 (s, 3H), 1.10 (s, 9H).

# EP 3 953 330 B1

Thr Gly Ile Ala Val Leu Asp Met Cys Ala Ser Leu Lys Glu Leu Leu  
260 265 270

5 Gln Asn Gly Met Asn Gly Arg Thr Ile Leu Gly Ser Ala Leu Leu Glu  
275 280 285

10 Asp Glu Phe Thr Pro Phe Asp Val Val Arg Gln Cys Ser Gly Val Thr  
290 295 300

Phe Gln  
305

15

## Claims

1. A compound of any one of formulae lh-1a, lh-1b, lh-1c, li-1a, li-1b, li-1c, lj-1a, lj-1b, lj-1c, lk-a, lk-b and lk-c:  
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